

IPL Cricket Analysis Dashboard Project Report

1. Project Title

IPL Cricket Analysis Dashboard using SQL Server and Power BI

2. Objective

The objective of this project is to analyze Indian Premier League (IPL) match data and provide interactive insights into team performance, player statistics, match results, powerplay analysis, and season-wise trends. The dashboard helps cricket enthusiasts, analysts, and decision-makers understand key patterns and performance metrics through data visualization.

3. Tools Used

Tool| Purpose

- Microsoft Excel| Initial data cleaning and preprocessing.
- SQL Server (SSMS)| Data storage, cleaning, transformation, and analytical queries.
- Power BI Desktop| Dashboard development and visualization
- AI | Background Images , Dax measure Analysis.

4. Approach / Methodology

Data Preparation

- Imported IPL datasets (Matches and Deliveries).
- Cleaned missing and duplicate records.
- Standardized team names and player information.
- Created SQL tables and loaded cleaned data.
- Developed SQL queries for player and team statistics.

Data Modeling in Power BI

- Connected Excel.csv files to Power BI.
- Established relationships between:
 - Matches → Deliveries (1:M)
 - Teams → Matches
 - Players → Deliveries

- Created DAX measures for KPIs and advanced calculations

5. Visualizations

Dashboard Pages

PAGE 1 - IPL OVERVIEW

KPI Cards for:

- Total Matches
 - Total Teams
 - Total Seasons
 - Total Runs
 - Total fours and sixes
-
- **Clustered Bar Chart for** - Matches Per Season
 - **Stacked Column Chart** - Team Wise Wins , Toss Decision Analysis
 - **Pie Chart** - Win Percentage for all Teams
 - **Slicer for Filtering** -Year

PAGE 2- TEAM ANALYSIS

- Win Percentage by Venue
 - Highest Team Score
 - Team Performance by Season
 - Lowest Team Score
-
- **Built Clustered Bar Chart** - Highest Team Score
 - **Built Pie Chart** - Team Performance by Season
 - **Built Donut Chart** - Win Percentage by Venue
 - **Built Stacked Bar Chart** - Lowest Team Score
 - **Used Slicer for filtering** - Team , Year, Venue

PAGE 3 - PLAYER ANALYSIS

- 1) **BATSMEN ANALYSIS**
- 2) **BOWLER ANALYSIS**

1) BATSMEN ANALYSIS

Top Run Scorer

- **Image Card** - Yellow Cap Holder

Built Cards For - Batsmen Name ,Total Runs,Strike Rate , Bounadary %, Fours, Sixes

Top 10 Batsmen Analysis

Used Image By Cloud Scope for - Batsmens Image Display

Built Cards For - Batsmen Name ,Total Runs, Strike Rate , Bounadary %, Fours, Sixes

Used Slicer for filtering - Top 10 Batsmen Names

2) BOWLER ANALYSIS

Top Wickets Taker

- **Image Card** - Red Cap Holder

Built Cards For - Bowler Name ,Total Wickets, Economy , Dot Ball%

➤ **Top 10 Bowler Analysis**

Used Image By Cloud Scope for - Bowler Image Display

Built Cards For - Bowler Name ,Total Wickets, Economy , Dot Ball%

Used Slicer for filtering - Top 10 Bowler Names

PAGE 4 – VENUE ANALYSIS

- Venue with Highest Score
- Best Chasing Ground
- Toss Impact by Stadium
- Average 1st Innings score

- **Built Clustered Column chart** - Venue with Highest Score
- **Built Line and clustered column chart** - Best chasing ground
- **Built Stacked Bar chart** - Toss Impact by Stadium
- **Built Tree Map** - Average 1st Innings Score
- **Used slicer for Filtering** - Venue and Year

PAGE 5 - MATCH INSIGHTS

1) POWERPLAY ANALYSIS

2) DEATHOVER ANALYSIS

- Winner by Toss decision
 - Away win by venue
 - Team Bat first win and team chase win
 - Most successful chase by team
- **Built Donut chart** - Winner by Toss decision
 - **Built Pie chart** - Away win by venue
 - **Built Stacked bar chart** - Team Bat first win and team chase win
 - **Built Clustered Bar chart** - Most successful chase by team
 - **Used Slicer for filtering** - Team

1) POWERPLAY ANALYSIS

- Powerplay Balls by bowler
- Overwise runs
- Powerplay runs by batter
- Powerplay wickets by bowler

KPI Cards for:

- Powerplay Runs
 - Powerplay Runrate
 - Powerplay Wickets
 - Powerplay Boundary %
 - Powerplay Dotball%
- **Built Clustered Bar chart** - Powerplay Balls by bowler
 - **Built Stacked Area chart** - Overwise runs
 - **Built Donut chart** - Powerplay runs by batter
 - **Built Funnel** - Powerplay wickets by bowler
 - **Used Slicer for Filtering** - Batter,Bowler

2) DEATHOVER ANALYSIS

KPI Cards:

- Deathover runs
- Deathover wickets
- Average Deathover RunRate
- Average Deathover Score
- Deathover Balls

- Deathover runs By Batter
- Deathover Strikerate by batter
- Deathover wickets by bowler
- Deathover Economy by bowler
- Deathover Economy and Batter RunRate by team

- **Built Clusterd Bar Chart** - Deathover runs By Batter
- **Built Funnel** - Deathover strikerate by batter
- **Built Piechart** - Deathover wickets by bowler
- **Built Linechart** - Deathover economy by bowler
- **Built Clustered Columnchart** - Deathover Economy and Batter RunRate by team
- **Used Slicer for Filtering** - Batter and Bowler

Interactive Features

- Team Slicer
- Season Slicer
- Venue Slicer
- Drill-through Analysis

6. Testing & Validation

- Validated SQL query results with Power BI visuals.
- Verified DAX calculations and KPIs.
- Tested slicers and cross-filter interactions.
- Ensured accuracy of team-wise and player-wise statistics.

7. Challenges Faced

- Handling large ball-by-ball IPL datasets.
- Managing relationships between multiple tables.
- Creating accurate DAX measures for advanced cricket metrics.
- Optimizing dashboard performance with multiple visuals.
- Maintaining consistent team names across seasons.

8. Conclusion

The IPL Cricket Analysis Dashboard provides comprehensive insights into team performance, player achievements, match outcomes, and powerplay trends. By combining SQL Server for data processing and Power BI for visualization, the dashboard delivers an interactive and data-driven platform for cricket analysis. The project demonstrates practical skills in SQL, data modeling, DAX, and business intelligence reporting.